Lab 3 Bash Scripting
Mohamed Sameh Elkholy

1)

Activities Terminal Mar 3 12:41 PM en 74 %

Open Bash.sh ~/Documents/GitHub/ITI-Assignments/Bash Scripting Save

```bash
#!/bin/bash
read -p "Enter a character: " ch
case $ch in
  [A-Z]) echo "Upper Case" ;;
  [a-z]) echo "Lower Case" ;;
  [0-9]) echo "Number" ;;
  "")    echo "Nothing" ;;
  *)     echo "Special character" ;;
esac
```

mohamed@Mohamed-Sameh-PC: ~/Documents/GitHub/ITI-A...

```
mohamed@Mohamed-Sameh-PC:~/Documents/GitHub/ITI-Assignments/Bash Scripting$ ./Ba
sh.sh
Enter a character: a
Lower Case
mohamed@Mohamed-Sameh-PC:~/Documents/GitHub/ITI-Assignments/Bash Scripting$ ./Ba
sh.sh
Enter a character: A
Upper Case
mohamed@Mohamed-Sameh-PC:~/Documents/GitHub/ITI-Assignments/Bash Scripting$ ./Ba
sh.sh
Enter a character: 1
Number
mohamed@Mohamed-Sameh-PC:~/Documents/GitHub/ITI-Assignments/Bash Scripting$ ./Ba
sh.sh
Enter a character: @
Special character
mohamed@Mohamed-Sameh-PC:~/Documents/GitHub/ITI-Assignments/Bash Scripting$
```

sh Tab Width: 8 Ln 9, Col 5 INS

2)

Activities Terminal Mar 3 12:49 PM en 81 %

Open Bash.sh ~/Documents/GitHub/ITI-Assignments/Bash Scripting Save

```bash
#!/bin/bash
read -p "Enter a string: " str
if [ -z "$str" ]; then
  echo "Nothing"
elif [[ "$str" =~ ^[A-Z]+$ ]]; then
  echo "Upper Cases"
elif [[ "$str" =~ ^[a-z]+$ ]]; then
  echo "Lower Cases"
elif [[ "$str" =~ ^[0-9]+$ ]]; then
  echo "Numbers"
else
  echo "Mix"
fi
```

mohamed@Mohamed-Sameh-PC: ~/Documents/GitHub/ITI-A...

```
mohamed@Mohamed-Sameh-PC:~/Documents/GitHub/ITI-Assignments/Bash Scripting$ ./Ba
sh.sh
Enter a string: A
Upper Cases
mohamed@Mohamed-Sameh-PC:~/Documents/GitHub/ITI-Assignments/Bash Scripting$ ./Ba
sh.sh
Enter a string: a
Lower Cases
mohamed@Mohamed-Sameh-PC:~/Documents/GitHub/ITI-Assignments/Bash Scripting$ ./Ba
sh.sh
Enter a string: 1
Numbers
mohamed@Mohamed-Sameh-PC:~/Documents/GitHub/ITI-Assignments/Bash Scripting$ ./Ba
sh.sh
Enter a string:
Nothing
mohamed@Mohamed-Sameh-PC:~/Documents/GitHub/ITI-Assignments/Bash Scripting$ ./Ba
sh.sh
Enter a string: A1
Mix
mohamed@Mohamed-Sameh-PC:~/Documents/GitHub/ITI-Assignments/Bash Scripting$
```

sh Tab Width: 8 Ln 13, Col 3 INS

3)

Activities Terminal Mar 3 1:15 PM en 99 %

Open Bash.sh ~/Documents/GitHub/ITI-Assignments/Bash Scripting Save

```
#!/bin/bash
for item in ~/*; do
  chmod +x "$item"
done
```

mohamed@Mohamed-Sameh-PC: ~/Documents/GitHub/ITI-A...

```
mohamed@Mohamed-Sameh-PC:~/Documents/GitHub/ITI-Assignments/Bash Scripting$ ./Ba
sh.sh
mohamed@Mohamed-Sameh-PC:~/Documents/GitHub/ITI-Assignments/Bash Scripting$
```

sh Tab Width: 8 Ln 4, Col 5 INS

4)

Activities Terminal Mar 3 12:55 PM en 87 %

Open Bash.sh ~/Documents/GitHub/ITI-Assignments/Bash Scripting Save

```
#!/bin/bash
mkdir -p ~/backup
for file in ~/*; do
  [ -f "$file" ] && cp "$file" ~/backup/
done
```

mohamed@Mohamed-Sameh-PC: ~/Documents/GitHub/ITI-A...

```
mohamed@Mohamed-Sameh-PC:~/Documents/GitHub/ITI-Assignments/Bash Scripting$ ./Ba
sh.sh
mohamed@Mohamed-Sameh-PC:~/Documents/GitHub/ITI-Assignments/Bash Scripting$
```

sh Tab Width: 8 Ln 5, Col 5 INS

5)

Activities Terminal Mar 3 1:11 PM en 99 %

Open Bash.sh ~/Documents/GitHub/ITI-Assignments/Bash Scripting Save

```bash
#!/bin/bash
for user in $(cut -d: -f1 /etc/passwd); do
  mail -s "Hello" "$user" < mtemplate
done
```

mohamed@Mohamed-Sameh-PC: ~/Documents/GitHub/ITI-A...

```
mohamed@Mohamed-Sameh-PC:~/Documents/GitHub/ITI-Assignments/Bash Scripting$ ./Ba
sh.sh
mohamed@Mohamed-Sameh-PC:~/Documents/GitHub/ITI-Assignments/Bash Scripting$ mail
"/var/mail/mohamed": 1 message 1 new
>N   1 Mohamed Sameh      Tue Mar  3 13:09  20/535   Hello
?
Return-Path: <mohamed@Mohamed-Sameh-PC>
X-Original-To: mohamed
Delivered-To: mohamed@Mohamed-Sameh-PC
Received: by Mohamed-Sameh-PC (Postfix, from userid 1000)
        id 163B6C2E49; Tue,  3 Mar 2026 13:09:58 +0200 (EET)
Subject: Hello
To: mohamed@Mohamed-Sameh-PC
User-Agent: mail (GNU Mailutils 3.14)
Date: Tue,  3 Mar 2026 13:09:58 +0200
Message-Id: <20260303110958.163B6C2E49@Mohamed-Sameh-PC>
From: Mohamed Sameh <mohamed@Mohamed-Sameh-PC>

Subject: Quick Note

Hey,

Hope you're doing well. Just wanted to check in.
```

sh Tab Width: 8 Ln 1, Col 12 INS

6)

Activities Terminal Mar 3 1:16 PM en 99 %

Open Bash.sh ~/Documents/GitHub/ITI-Assignments/Bash Scripting Save

```bash
#!/bin/bash
while true; do
  mail_file="/var/mail/$USER"
  [ -s "$mail_file" ] && echo "You have new mail!" || echo "No new mail."
  sleep 10
done
```

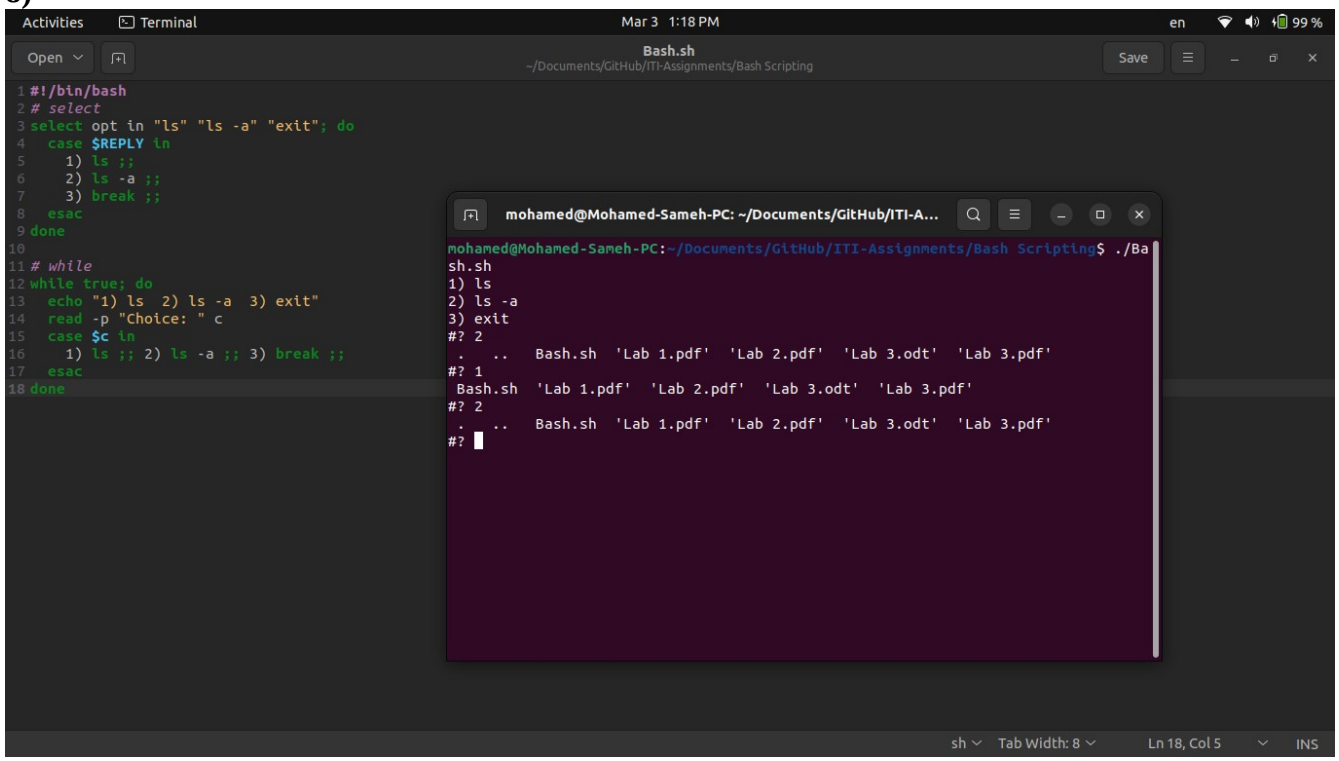
mohamed@Mohamed-Sameh-PC: ~/Documents/GitHub/ITI-A...

```
mohamed@Mohamed-Sameh-PC:~/Documents/GitHub/ITI-Assignments/Bash Scripting$ ./Ba
sh.sh
No new mail.
No new mail.
No new mail.
No new mail.
No new mail.
```

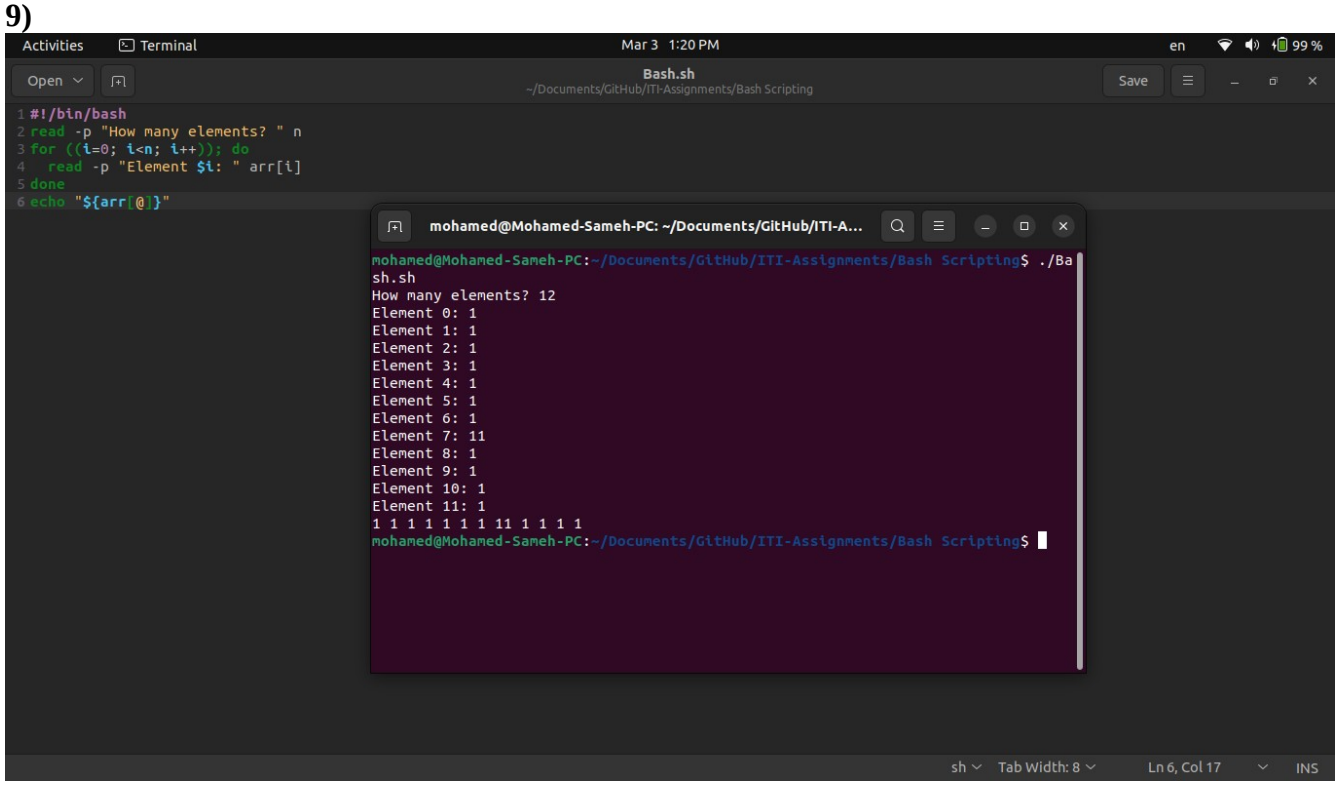
sh Tab Width: 8 Ln 6, Col 5 INS

7) The output is **1**

8)



9)



10)

Activities Terminal Mar 3 1:20 PM en 99 %

Open Bash.sh ~/Documents/GitHub/ITI-Assignments/Bash Scripting Save

```bash
#!/bin/bash
read -p "How many numbers? " n
sum=0
for ((i=0; i<n; i++)); do
  read -p "Enter number: " nums[i]
  sum=$((sum + nums[i]))
done
echo "Average: $((sum / n))"
```

mohamed@Mohamed-Sameh-PC: ~/Documents/GitHub/ITI-A...

```
mohamed@Mohamed-Sameh-PC:~/Documents/GitHub/ITI-Assignments/Bash Scripting$ ./Ba
sh.sh
How many numbers? 5
Enter number: 1
Enter number: 2
Enter number: 3
Enter number: 4
Enter number: 5
Average: 3
mohamed@Mohamed-Sameh-PC:~/Documents/GitHub/ITI-Assignments/Bash Scripting$
```

sh Tab Width: 8 Ln 8, Col 29 INS

11)

Activities Terminal Mar 3 1:21 PM en 99 %

Open Bash.sh ~/Documents/GitHub/ITI-Assignments/Bash Scripting Save

```bash
#!/bin/bash
mysq() {
  echo $(( $1 * $1 ))
}
read -p "Enter number: " x
mysq $x
```

mohamed@Mohamed-Sameh-PC: ~/Documents/GitHub/ITI-A...

```
mohamed@Mohamed-Sameh-PC:~/Documents/GitHub/ITI-Assignments/Bash Scripting$ ./Ba
sh.sh
Enter number: 12
144
mohamed@Mohamed-Sameh-PC:~/Documents/GitHub/ITI-Assignments/Bash Scripting$
```

sh Tab Width: 8 Ln 6, Col 8 INS